Ex. No :08 **App Architecture – UI Layer** **Jason Freddy F P**

Aim

To understand the UI layer in Android architecture by handling user interaction and updating UI using Activity and ViewModel.

Procedure

1 Open Android Studio.

. Create a new project (Empty Activity, Kotlin).

2 Design UI in activity_main.xml using:

. TextView (to display message)

3 Button (to trigger action)

.4 Create a **ViewModel class** to manage data.

. Open MainActivity.kt.

5 Connect UI components using findViewById.

. Use ViewModel to fetch data.

6 Handle button click and update UI.

. Run the app and observe output.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="20dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/tvTitle"
      android:text="UI Layer Demo"
      android:textSize="22sp"
      android:gravity="center"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>

   <TextView
      android:id="@+id/tvMessage"
      android:text="Message will appear here"
      android:textSize="18sp"
      android:paddingTop="20dp"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>
```

```
    <Button
        android:id="@+id/btnLoad"
        android:text="Load Data"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainViewModel.kt

```
package com.example.uilayerdemo

import androidx.lifecycle.ViewModel

class MainViewModel : ViewModel() {

    fun getMessage(): String {
        return "Hello from ViewModel"
    }
}
```

MainActivity.kt

```
package com.example.uilayerdemo


import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.activity.viewModels
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private val viewModel: MainViewModel by viewModels()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvMessage = findViewById<TextView>(R.id.tvMessage)
        val btnLoad = findViewById<Button>(R.id.btnLoad)

        btnLoad.setOnClickListener {
            tvMessage.text = viewModel.getMessage()
        }
    }
}
```

Result

The application successfully:

- Handles user interaction through UI layer
- Uses ViewModel to provide data

- Displays dynamic message on button click